Harsha Ganugula

pavanharsha412@gmail.com | (940)843-3049 | LinkedIn | GitHub | Portfolio | USA

Summary

Software Developer with 3+ years of experience building scalable full-stack applications, with a strong focus on modern, responsive front-end development using React and JavaScript (ES6+). Experienced in designing intuitive user interfaces, role-based dashboards, and real-time features while supporting them with robust backend systems built on Java (Spring Boot) and Python. Skilled in RESTful APIs, event-driven microservices, and cloud-native deployment on AWS and Azure. Passionate about creating seamless user experiences backed by reliable, high-performance architecture.

Technical Skills

- **Frontend**: React.js, TypeScript, JavaScript (ES6+), Redux, React Hooks, HTML5, CSS3, Responsive Design, WebSocket's
- **Backend**: Java (Spring Boot, Spring Security), Python (Django REST Framework), RESTful APIs, Microservices Architecture, Apache Kafka, JWT Authentication
- **Data bases**: PostgreSQL, MySQL, MongoDB, AWS RDS
- **Dev Ops & Cloud**'s: AWS (Elastic Beanstalk, CodePipeline, CloudWatch, Auto Scaling), Microsoft Azure (App Services, Blob Storage), Jenkins, GitHub Actions, CI/CD Pipelines
- **Testing**: JUnit, Mockito, PyTest, Jest, Embedded Kafka, Unit & Integration Testing
- **Tools & Collaboration**: Git, Jira, Agile/Scrum, Cross-Functional Collaboration

Professional Experience

Software Engineer, BlueCross BlueShield 01/2024–Present | Remote, USA

- Built scalable, role-based dashboards and workflow interfaces using React, TypeScript, and modern JavaScript (ES6+), improving user interaction efficiency and streamlining internal processes.
- Developed reusable, strongly typed UI components using React Hooks and TypeScript interfaces, enhancing code maintainability, reducing runtime errors, and enabling consistent frontend architecture.
- Optimized frontend performance and API integration, improving data rendering efficiency and ensuring a smooth user experience across enterprise applications.
- Integrated TypeScript-based frontend applications with backend microservices, enabling type-safe API communication, robust error handling, and efficient data flow.
- Designed and developed secure RESTful APIs using Spring Boot to support eligibility, document processing, and workflow services, improving response time by 30%.
- Implemented JWT-based authentication and role-based access control (RBAC) using Spring Security, strengthening application security for internal and customer-facing systems.
- Migrated real-time communication from WebSocket's to Apache Kafka, improving scalability, fault tolerance, and event-driven processing capabilities.
- Built Kafka producers and consumers to support asynchronous workflows, enabling real-time document tracking and status updates.
- Enhanced system monitoring and observability using AWS CloudWatch, reducing production issue diagnosis time by 25% and improving reliability.
- Improved code quality by implementing unit and integration tests using JUnit, Mockito, and Embedded Kafka, achieving 85%+ test coverage.
- Collaborated with cross-functional teams including product managers, designers, QA, and DevOps to deliver scalable applications in Agile environments

Software Developer, Cisco 06/2021–08/2022 | Bangalore, India

- Developed responsive user interfaces using Java, Spring Boot, and Thymeleaf, implementing contract upload, approval workflows, and role based access features.
- Designed and built backend REST APIs using Python and Django REST Framework to support enterprise-scale document processing and approval systems.
- Implemented frontend validation logic and optimized UI rendering performance, improving user workflow speed and reducing manual processing errors.
- Designed and optimized PostgreSQL database schemas using indexing and normalization strategies, reducing query latency by 45% and improving system efficiency.
- Integrated Jenkins CI pipelines to automate builds, testing, and deployment processes, accelerating release cycles by 50%.
- Built event-driven backend components to process contract lifecycle updates and notification triggers across distributed modules.
- Developed NLP-based clause extraction and classification features using custom-trained spaCy models, achieving 92% accuracy in automated document analysis.
- Implemented automated unit and integration tests using PyTest and JUnit, maintaining 90% code coverage and reducing production defects.

- Integrated JWT-based authentication and secure authorization mechanisms to ensure compliance with internal enterprise security standards.
- Deployed applications to Microsoft Azure using App Services and CI/CD pipelines, achieving 99.9% uptime and improving deployment reliability.
- Collaborated with cross-functional stakeholders to gather requirements, refine features, and deliver consistent sprint outcomes under Agile methodology.
- Contributed to code reviews and architectural discussions to maintain clean code standards and scalable design patterns.

Education

Master of Science in Information System and Technologies 08/2022–05/2024
University of North Texas **Denton, TX, USA**

Projects

SaaS Analytics Dashboard – Production-Style Admin Panel

React.js, REST API Integration, Data Visualization, Client-Side Pagination, CSS Variables | GitHub | Live Demo

- Developed a modern SaaS-style analytics dashboard simulating a production admin panel with clean component architecture and reusable UI modules.
- Integrated live user data from a public REST API using fetch inside useEffect, implementing loading states, error handling, and conditional rendering for resilience.
- Implemented client-side search and pagination logic using derived state and array slicing, ensuring smooth and consistent user experience.
- Built reusable components including Sidebar, Header, Metric Cards, Charts, and Activity Tables to promote scalability and maintainable UI systems.
- Designed dark/light theme switching using CSS variables and root-level class toggling for dynamic UI updates without page reloads.
- Optimized rendering performance and state updates to efficiently handle larger datasets and minimize unnecessary re-renders.
- Structured the project into modular folders (components, pages, styles, data) to reflect production-grade frontend organization.
- Deployed to Vercel with optimized build configuration to ensure fast load times and consistent performance.

SubTrack – SaaS Subscription Management Platform

React.js, Context API, React Router, Local Storage, SaaS Billing Logic | GitHub | Live Demo

- Built a SaaS-style subscription management application that models real-world billing cycles and calculates normalized Monthly Recurring Revenue (MRR) for monthly and yearly plans.
- Implemented global state management using React Context API (AuthContext and SubscriptionContext) to centralize authentication and subscription logic while avoiding prop drilling.
- Developed feature gating logic to simulate Free vs Pro subscription tiers, restricting usage dynamically and enforcing business rules at the UI layer.
- Implemented protected routing using React Router with conditional rendering to prevent unauthorized dashboard access.
- Persisted application state using Local Storage to simulate backend data storage in a frontend-only MVP environment.
- Structured the application using reusable components and modular architecture to support scalability and maintainability.
- Designed responsive UI components with intuitive dashboards to provide real-time subscription analytics and expense visibility.
- Deployed the application to Vercel with optimized production builds and environment configuration.